



Harendra Kumar Dadhich

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DoB:05/10/1988

Github Profile:<https://github.com/harendradadhich>

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Kaggle Profile:<https://www.kaggle.com/harendra123>

I am a highly skilled and experienced **Geospatial Data Scientist** with a proven track record of developing and implementing advanced machine learning and deep learning algorithms for geospatial data analysis. Seeking to leverage 12+years of experience to contribute effectively to esteemed organizations in driving innovation and data-driven decision-making processes.

SKILLSET

Data Science/Analytics
Predictive Modeling,
Decision Making,
AI, Geospatial,
LLM, Spark, Hadoop,
Remote Sensing
Python, Machine
Learning, Text
Mining/NLP Modeling &
Simulation Global
Positioning System ,
Crop Classification,
Databricks, GitHub,
Crustal deformation
Project Execution
Satellite Image
Processing,
QGIS, Project
Coordination

PROFILESUMMARY

- **B.Tech.(ECE)** with total 12+ years of experience and over 7+ years of experience in Data Analytics, Data Science, Machine learning & Predictive Modeling implementation and Analysis.
- Strong analytical skill with knowledge of implementation and analysis of **Machine Learning** models like **Regression** (Linear, Multiple, logistic etc.), **Classification** (KNN, Random Forest), **Clustering**, **Natural Language Processing (NLP)**, **Large Language Model (LLM)**.
- Strong analytical skills with Knowledge of implementation and analysis of **Deep learning models** like **Convolution Neural Network (CNN)**, **Recurrent Neural Network (RNN)**, **LSTM**, **ANN** etc.
- Developed AI-driven solutions using **Lang chain**, leveraging advanced NLP techniques to enhance automation and decision-making processes.
- Keen planner, strategist & implementer with excellence in ensuring smooth functioning of overall operations; enhancing operational efficiency by making use of leading-edge technologies including business reviews
- Successfully completed projects in **Remote Sensing** on Satellite Meteorology and Global Positioning System (GPS) Technology for **crustal deformation** mapping.
- Possess knowledge of **Python Programming** (Data Science/Analytics).
- Achievement oriented professional with strong decision-making and **Problem-solving skills** for enabling effective solutions with excellent communication skills.

ITSKILLS

Operating Systems:	Linux (RedHat, Ubuntu) and Windows(9x,XP,7,8&10)
Languages & Framework:	Python , ShellScripting, PySpark ,C/C++(Basic)and OOPs,Pytorch,SKlearn,Tensorflow,Keras,PySpark
Data Science:	Regression,Classification,Clustering, MachineLearning ,TextMining(NLP) ,Computer Vision, CNN, Deep Learning, , XGboost , Pytorch.
Gen AI:	Open AI LLM Models, FLAN-T5, Langchain, SAM, ASR models, opensource hugging face models
Tools:	Anaconda,Jupyter Notebook,PowerBI(Visualization tool),Tableau, Git, Data Bricks
S/W:	ArcGIS (Desktop),GAMIT/GLOBK,SURFER,GRAPHER,GMT,GMT-SAR, BEAM , SNAP , QGIS
Cloud Platform:	AWS , ECR, Sagemaker , S3, EC2, GCP, Big Query , Google Dataflow etc.

ORGANISATIONAL EXPERIENCE

LTIMindtree

(Aug'24–till date)

- Developing predictive models for crop classification in India using AI & ML.
- Working on various geospatial use case of AI and ML in Geospatial Industry.

- Making AI-ML solutions for projects related to Geospatial domain.

BirlaSoft

(May'22–Aug'24)

Data Scientist

- **Developing predictive models** and applying machine learning algorithms to **extract insights and drive data- based decision-making**
- Creating comprehensive reports and dynamic dashboards utilizing **PowerBI/Tableau** for data visualization and analysis
- ETL task using **AWSRedshift** and save the data in **AamazonS3 Buckets**.
- Data Analysis: Expert in exploring and analysis the geospatial data like satellite data, drone data to identify patterns and extracting valuable insights using statistical techniques and visualization.

National Center for Seismology, Ministry of Earth Science, New Delhi

(Dec'14 -Mar' 22)

Data Scientist (RA)

- Crustal deformation mapping using **Global Positioning System(GPS) and SAR data**
- Conducted spatial data analysis and visualization to identify patterns, trends, and relationships.
- Developed and implemented state-of-the-art **machine learning and deep learning algorithms for geospatial data processing and analysis**.
- Earthquake Intensity map generation
- Worked on **Landslide detection and prediction using Machine learning** and satellite data, Digital elevation models (DEM), DSM etc.
- Designed and optimized geospatial models and algorithms to support decision-making processes.
- Estimating refined **Magnitude of Earthquake using Machine Learning Models E.g. Linear Regression**
- Detection and Prediction of Earthquake using **Machine Learning in Seismology field**.
- Satellite based forecast model based on extrapolation technique and regression equations
- Participated actively in regular review meetings, fostering knowledge sharing and continuous improvement
- Conducted detail analysis of forecast model and **cross validation** of the prediction result with actual observation: compared models with other forecast techniques that use other data mining approaches (**like Logistic Regression, Forecasting and so on**)
- Mentored MTech. Tech. and MSc. Students for projects & internships in NCS mainly on the topic on Data Mining and Machine Learning, Scripting in Python, Shell Scripting, Data Validation

International Center for Radio Science, Jodhpur

(Feb'11-Jan'14)

Junior Research Fellow

- Tasks which involved the use advanced modelling techniques – Regression, Clustering & Timeseries to provide actionable results
- Oceanography, **Agricultural**, Seismological and Atmospheric applications of **Microwave Remote Sensing**.
- Detection of **Cyclones, Oil spills, Tsunami using active &passive sensors onboard satellite data**
- Processed the data and applied data mining to Seismological Data consolidation problem
- Worked for the estimation of **Soil Moisture using SMOS satellite**
- Worked for the retrieval of **ocean salinity using SMOS satellite**
- Instituted Timeseries, Clustering & Extrapolation model to maintenance records to predict crustal deformation

EDUCATION QUALIFICATIONS

- Pursuing M.Tech in DataScience from Birla Institute of Technology & Science, (BITS), Pilani.
- B.Tech(Electronics & Communication) in 2010 from Jodhpur Engineering College Research Centre, Rajasthan Technical University, Kota with 64.47%.
- Intermediate in 2006 from R.B.S.E board with 73.54%.
- Matriculation in 2004 from R.B.S.E board with 75.00%.

INDUSTRIAL TRAININGS & CERTIFICATIONS

- Data Science and Machine Learning using Python from Analytix Labs, Gurgaon.
- ArcGIS1: Introduction to GIS from ESRI India

- **AWSPartner: Machine Learning on AWS(Technical)(Classroom)**
- Machine Learning from Solo learn
- Prepare,Clean,Transform,andLoadDatausingPowerBIbyCoursera
- Calculus for Machine Learning and Data Science by Deep learning from Coursera
- MachineLearning: Regression by University of Washington from Coursera
- Introduction to Generative AI by Google Cloud from Coursera
- AI Agents and Agentic AI with Python & Generative AI from Vanderbilt University
- Introduction to Retrieval Augmented Generation (RAG) from Duck University
- Transformer Models and BERT Model from Google Cloud
- LangChain Chat with Your Data from Deeplearning.AI

PROFESSIONAL SKILLS

- **Strong analytical and problem-solving skills**
- Utilized **Remote Sensing data** for monitoring **soil moisture** levels and detecting changes in terrain topography.
- Developed custom algorithms for processing **multi spectral imagery** to detect vegetation stress and assess crop health.
- **Data Analysis:** Expert in exploring and analyzing complex datasets, identifying patterns, and extracting valuable insights using statistical techniques and visualization tools.
- **Machine Learning:** Proficient in developing and implementing machine learning algorithms and models, including **regression, classification, clustering, and deep learning**, using frameworks such as TensorFlow and scikit-learn.
- **Perform explanatory data analysis on huge datasets**
- **Statistical Modeling:** Extensive experience in applying statistical techniques, hypothesis testing, and experimental design to solve business problems and optimize processes.
- **Programming:** Proficient in Python programming language, with a strong understanding of data manipulation, cleaning, and preprocessing techniques.
- **Data Visualization:** Skilled in creating compelling visualizations and dashboards using tools seaborn and matplotlib to effectively communicate insights and findings to stakeholders.
- **Team Leadership:** Demonstrated leadership skills in managing and collaborating with cross-functional teams, providing guidance, and fostering a collaborative and innovative work environment.
- Strong experience in applying Machine learning algorithms linear regression, Logistic regression, Naive Bayes, KNN, Decision trees, Random forest and SVM. Visualization libs **Matplotlib, Seaborn**
- Good knowledge in segregating the data points by creating clusters using k-means algorithm. Having in depth knowledge in **feature selection** and dimensionality reduction using unsupervised technique (Principal Component Analysis) Good at Feature Engineering, Outlier detection using IQR box plots.
- **NLP Expertise:** Extensive experience in applying NLP techniques and algorithms to process, analyze, and extract insights from textual data.
- **Sentiment Analysis:** Proficient in performing sentiment analysis on large-scale text data, enabling sentiment-based decision-making and customer feedback analysis.
- **Text Classification:** Strong background in developing text classification models for various applications, including topic classification and sentiment classification.
- **Text Preprocessing:** Strong knowledge of text preprocessing techniques, including tokenization ,stemming, lemmatization, and stop-word removal, to prepare text data for analysis and modeling.
- **NLP Libraries and Tools** Proficient in utilizing popular NLP libraries and frameworks such as NLTK, spaCy, and Gensim.

PROOFOFCONCEPTS(MachineLearning)

- Predict Credit Consumption of Customer For Leading Bank
- Cyber Security–Network Security: Network Intrusions Detection Case Study
- Electricity Demand Estimation For Leading Electricity Distributor
- Marketing Insights For E-Commerce Company

- Customer Review Analysis For Leading woman clothing E-Commerce Company

RESEARCH PAPER ACCPETED AND PUBLISHED

- **Dadhich, H.K. (2023).** "Overview of Artificial Intelligence (AI) and Machine Learning (ML) in Seismology".published in Geo hazards: Analysis, Modelling, and Forecasting, Advances in Natural and Technological Hazards Research, vol 53, **2023, Springer, Singapore.** https://doi.org/10.1007/978-981-99-3955-8_12
- Vandana, **Harendra K. Dadhich**, Himanshu Mittal, O.P. Mishra, "Ground motion prediction equation for NW Himalaya and its surrounding region,"Quaternary Science Advances,**2023**,100136,ISSN 2666-0334,<https://doi.org/10.1016/j.qsa.2023.100136>.
- Kapil Mohan, R Gautam, S. Dhamodharan, **H K Dadhich**, Prasanta Chingtham, Kalachand Sain, O. P. Mishra, "Geoelectric Characterisation of the Junction of seismically active Delhi Hardwar Ridge and Delhi Sargodha Ridge" published in **Scientific reports, Nature, 2023**, Sci Rep 13, 18488 (2023). <https://doi.org/10.1038/s41598-023-42722>
- **Harendra K Dadhich**, Sanjay K Prajapati, Sumer Chopra, Babita Sharma "Development of Empirical Relationships between Ground-Motion Parameters and Modified Mercalli Intensity for Nepal Region" presented in 30thHimalaya-Karakoram- Tibet Workshop held in Wadia Institute of Himalayan Geology, Dehradun,2015.
- O P N Calla, **Harendra Kumar Dadhich**, Shruti Singhal,"Oil Spill Detection Using Multi Frequency Microwave Sensor Onboard Satellite; SSM/I AND AMSR-E", published in, 2013 IEEE -IGARSS Melbourne, VIC, Australia, pp3710-3713. [doi:10.1109/IGARSS.2013.6723636](https://doi.org/10.1109/IGARSS.2013.6723636)
- O P N Calla, **Harendra Kumar Dadhich**, Shruti Singhal, PriyankaToshniwal"StudyofRelationship between BrightnessTemperatureandWindSpeed over Indian Ocean",Oral Presentation in International Conference on Microwaves, Antenna, Propagation **and Remote Sensing** (ICMARS) 2012 ,pp 125-127.(3rdPrizeAwarded)
- OPNCalla,**HarendraKumarDadhich**,ShrutiSinghal,"ComparativeStudyofSeaSurfaceSalinityobtain from Aquarius/SAC-D Mission and ArgoFloats over Indian Ocean", presented as Poster presentation in SMOS and AquariusWorkshop2013, IFREMER, Brest, France, 15- 17April,2013,p.39.

PAPER PUBLICATIONS

- VKGahalaut, StaceyS Martin,D Srinagesh, SL Kapil,GSuresh,SauravSaikia,VikasKumar, **Harendra Dadhich**, Aqeel Patel, Sanjay K Prajapati, HP Shukla, JL Gautam, PR Baidya, Saroj Mandal, Ashish Jain,"Seismological, geodetic, macroseismic and historical context of the 2016 Mw 6.7 Tamenglong (Manipur) India earthquake", published in Tectonophysics, Volume 688, 12 October2016, Pages36-48.[doi:https://doi.org/10.1016/j.tecto.2016.09.017](https://doi.org/10.1016/j.tecto.2016.09.017)
- **Harendra Dadhich**, MS Naidu, Vineet K Gahalaut, JK Catherine, Amit K Bansal, V Rajeshwar Rao ,"A Glimpse of Crustal Deformation Through Earthquake Supercycle in the Andaman Region Using GPS Measurements," published in, The Andaman Islands and Adjoining Offshore: Geology, Tectonics and Palaeoclimate, 223-236, Springer, Cham, 2020.DOI:https://doi.org/10.1007/978-3-030-39843-9_11
- Sanjay K Prajapati, **Harendra K Dadhich**, Sumer Chopra," Iseismal map of the2015Nepal earthquake and its relationships with ground-motion parameters, distance and magnitude" published in Journal ofAsian Earth Sciences,Volume133, 1 January2017,Pages24–37
- OPNCalla, ShrutiSinghal,**HarendraKumarDadhich**"DetectionofTropicalCyclonePHETusing 19.35GHz SSM/I Data",published in Indian Journal of Geo Marine sciences, vol 44(3),March 2015, pp404-410.
- O P N Calla, **Harendra Kumar Dadhich**, Shruti Singhal "Oil spill detection using satellite data over Bombay High location in Arabian Sea", published in Indian Journal of Radio and Space physics, Vol42, February 2013, pp52- 59.

PROFESSIONALMEMBERSHIPANDPOSITIONSOFRESPONSIBILITIES

- Joint Secretary in 9th International Conference on Microwaves Antenna Propagation & remote sensing (ICMARS-2013)
- Treasurer of EXECUTIVE COUNCIL of IMS Jodhpur Chapter.
- IEEE GRSS Member (MembershipNumber92335456)

- IMS Life Member (MembershipNumberLM2436)
- Reviewer for IGARSS, IEEE and Journal of Geophysical Research: Machine Learning and Computation

DECLARATION

I hereby declare that the above-furnished information is correct up to best of my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.

Date:

Place:

(HarendraKDadhich)